Supplementary Material

SUPPLEMENTARY MATERIAL OVERVIEW

This supplementary material contains Supplementary Figures S1-S3 and complete embedded display versions of Supplementary Tables 1-11. The original CSV files, the combined Excel workbook and the figure source-data folder are included with the review package as machine-readable source files.

SUPPLEMENTARY FIGURES

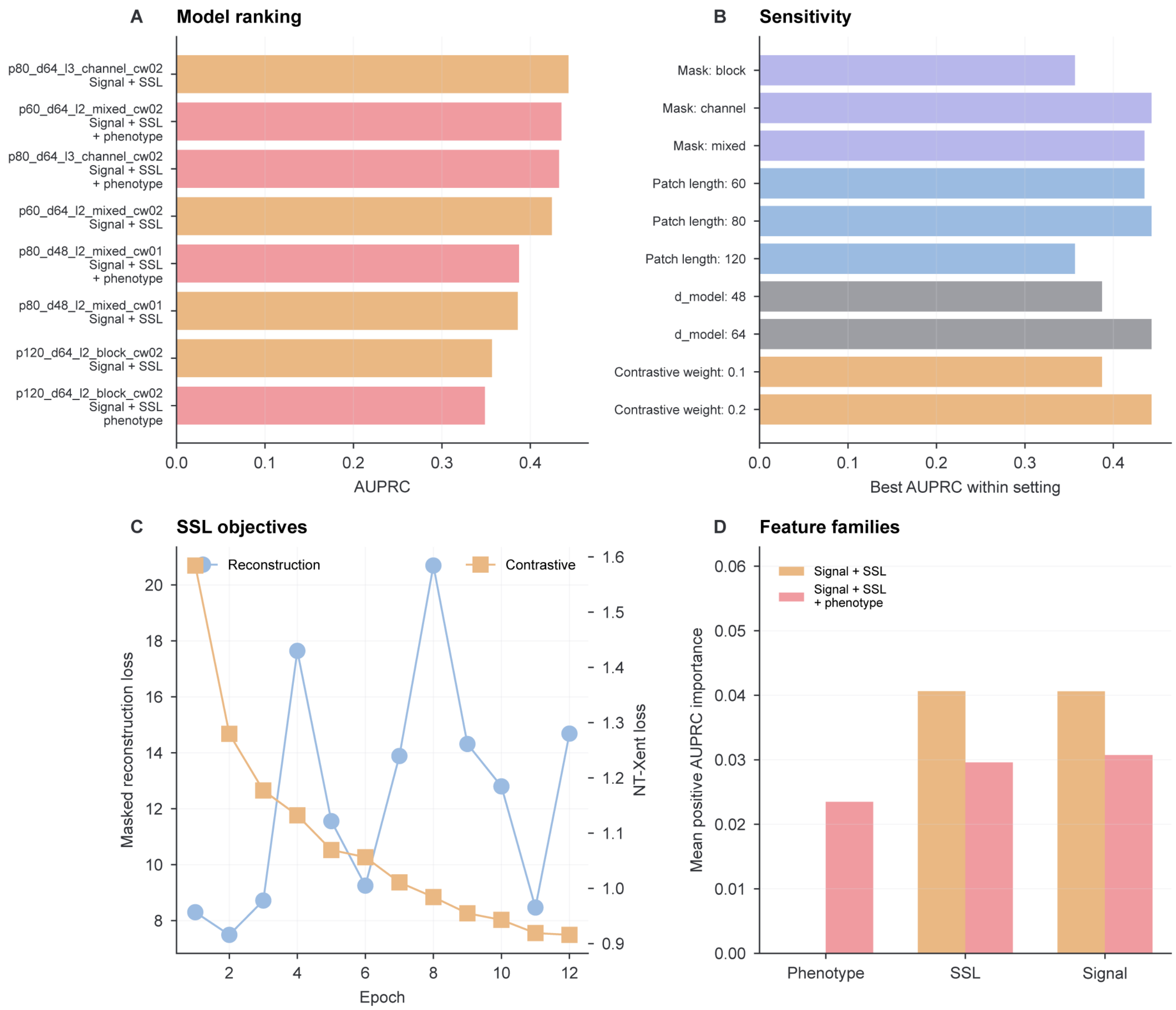

Figure S1. AI model-selection and sensitivity panels for the transformer phenotyping workflow. Supplementary analyses of the self-supervised AI method layer. (A) Exploratory transformer configuration and feature-set combinations ranked by held-out AUPRC. (B) Sensitivity summary across masking strategy, patch length, model dimension and contrastive objective weight, using the best AUPRC observed within each setting. Because these panels use held-out performance for method exploration, they should not be interpreted as a locked independent model-selection experiment. (C) Training-objective trace for a selected-architecture refit, showing masked reconstruction loss and NT-Xent contrastive loss across epochs; the oscillating reconstruction trace was used for training-dynamics inspection only and did not determine downstream representation selection. (D) Feature-family contribution to AUPRC-based permutation importance for signal-plus-SSL and final phenotype-augmented models.

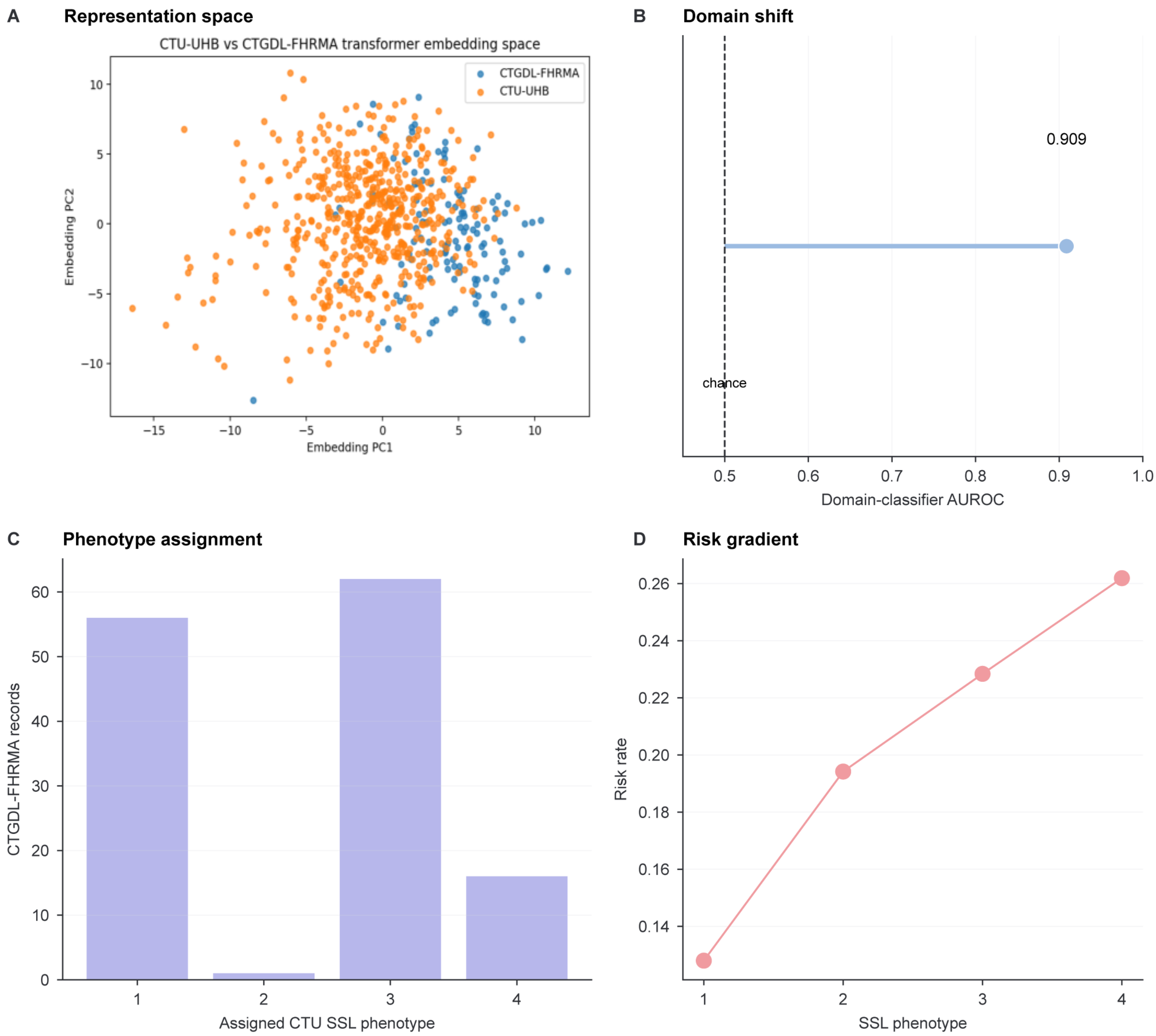

Figure S2. CTGDL-FHRMA supports external representation projection and domain-shift assessment. External representation analysis using CTGDL-FHRMA records. Because external neonatal outcome labels were not used for CTGDL-FHRMA, this analysis is interpreted as external representation projection and domain-shift assessment rather than external outcome validation. (A) Principal-component visualization of CTU-UHB and CTGDL-FHRMA transformer embeddings. (B) Domain separability quantified by a CTU-versus-CTGDL classifier. (C) CTGDL-FHRMA records assigned to CTU-derived SSL phenotypes. (D) CTU-UHB neonatal risk gradient across SSL phenotypes.

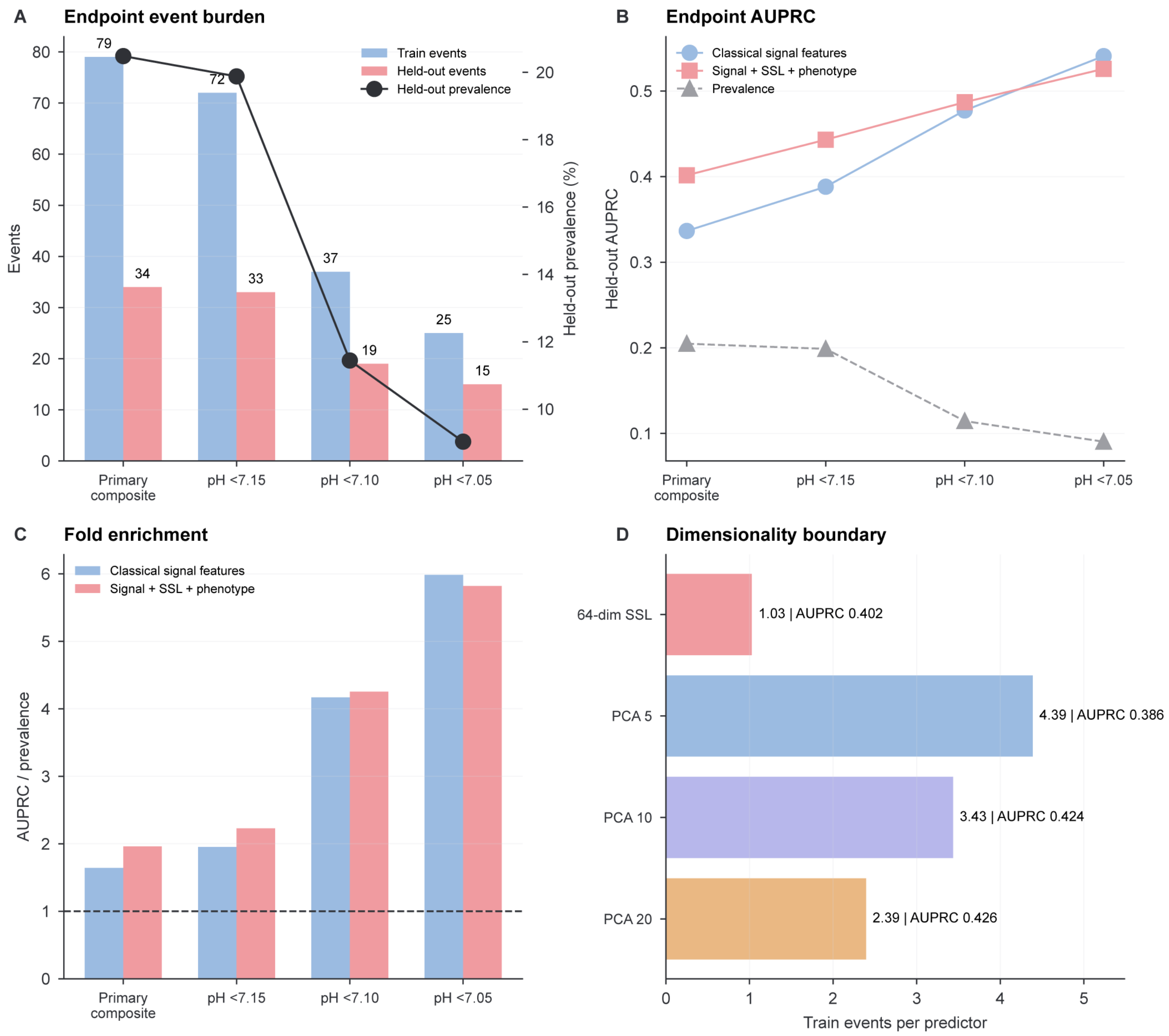

Figure S3. Endpoint-threshold and sample-size sensitivity analyses. Internal sensitivity analyses for endpoint definition and labelled-sample constraints. (A) Train and held-out event counts with held-out prevalence for the primary pH <7.15/Apgar composite and pH-only thresholds. (B) Held-out AUPRC across endpoint definitions for classical signal features and the final signal-plus-SSL-plus-signal-phenotype model, with endpoint prevalence shown as the random-classifier reference. (C) Fold enrichment over endpoint prevalence. (D) Train-event-per-predictor sensitivity for the full 64-dimensional SSL representation and PCA-compressed SSL summaries. These panels are internal sensitivity analyses and do not constitute external outcome validation or a severe-acidemia primary claim.

SUPPLEMENTARY TABLES

Supplementary Table 1. Dataset sources, cohort scale, record-level split, window counts, and external assessment role

Complete source table: Supplementary_Table_1_datasets_and_cohort_scale.csv. Rows: 8; columns: 10. The display is split into source-defined sub-tables and column blocks to preserve a readable table layout; no non-empty source rows or columns are omitted.

Table S1: Supplementary Table 1, column block 1 of 3

dataset or subset	role	records	windows	neonatal risk rate
CTU-UHB complete cohort	primary raw CTG cohort	552	7602	0.2047
CTU-UHB training split	model development	386	5301	0.2047
CTU-UHB held-out test split	internal validation	166	2301	0.2048
CTGDL-FHRMA	external representation/ domain-shift assessment	135	2604	
CTU-UHB / CTU-CHB Intrapartum CTG Database	primary raw signal cohort			
UCI Cardiotocography	feature-level baseline			
CTGDL	multi-source external/ harmonized resource			
CTU-CHB annotation dataset	optional expert annotation layer			

Table S2: Supplementary Table 1, column block 2 of 3

dataset or subset	role	notes	url	access
CTU-UHB complete cohort	primary raw CTG cohort	Open intrapartum FHR and uterine contraction signals with clinical metadata.		
CTU-UHB training split	model development	Record-level split to avoid window leakage.		
CTU-UHB held-out test split	internal validation	Held-out record-level test set for manuscript metrics.		
CTGDL-FHRMA	external representation/ domain-shift assessment	Used for embedding transport and phenotype assignment, not external outcome validation.		
CTU-UHB / CTU-CHB Intrapartum CTG Database	primary raw signal cohort	552 intrapartum CTG records with FHR/UC signals and clinical metadata	https://physionet.org/content/ctu-uhb-ctgdb/1.0.0/	open access
UCI Cardiotocography	feature-level baseline	2126 CTG feature rows; no raw signal	https://archive-beta.ics.uci.edu/dataset/193/cardiotocography	open access
CTGDL	multi-source external/ harmonized resource	Includes CTU-UHB derived files and other sources; SPAM subset notes DUA requirement	https://zenodo.org/records/18023488	open record with subset-specific caveats
CTU-CHB annotation dataset	optional expert annotation layer	Useful for clinically interpretable event annotation if files are accessible	https://doi.org/10.1016/j.dib.2020.105690	article/open supplementary source

Table S3: Supplementary Table 1, column block 3 of 3

dataset or subset	role	license or terms	status
CTU-UHB complete cohort	primary raw CTG cohort		
CTU-UHB training split	model development		
CTU-UHB held-out test split	internal validation		
CTGDL-FHRMA	external representation/domain-shift assessment		
CTU-UHB / CTU-CHB Intrapartum CTG Database	primary raw signal cohort	Open Data Commons Attribution License v1.0	verified
UCI Cardiotocography	feature-level baseline	UCI repository terms	verified
CTGDL	multi-source external/ harmonized resource	mixed; check each subset	verified with caveats
CTU-CHB annotation dataset	optional expert annotation layer	check publisher/data terms	identified

Supplementary Table 2. Held-out model performance with 2000-bootstrap confidence intervals

Complete source table: Supplementary_Table_2_model_performance_bootstrap.csv. Rows: 40; columns: 8. The display is split into source-defined sub-tables and column blocks to preserve a readable table layout; no non-empty source rows or columns are omitted.

Table S4: Supplementary Table 2, column block 1 of 2

model	model label	metric	point	bootstrap mean
Classical signal features	Signal features	auroc	0.6303	0.6293
Classical signal features	Signal features	auprc	0.3366	0.3522
Classical signal features	Signal features	brier	0.2525	0.2528
Classical signal features	Signal features	Calibration intercept	-1.384	
Classical signal features	Signal features	Calibration slope	0.4876	
SSL embeddings	SSL embeddings	auroc	0.4947	0.4936
SSL embeddings	SSL embeddings	auprc	0.2125	0.2269
SSL embeddings	SSL embeddings	brier	0.2639	0.2639
SSL embeddings	SSL embeddings	Calibration intercept	-1.361	
SSL embeddings	SSL embeddings	Calibration slope	-0.0415	
Signal + SSL	Signal + SSL	auroc	0.6025	0.6007
Signal + SSL	Signal + SSL	auprc	0.3765	0.3832
Signal + SSL	Signal + SSL	brier	0.2701	0.2707
Signal + SSL	Signal + SSL	Calibration intercept	-1.36	
Signal + SSL	Signal + SSL	Calibration slope	0.3136	
signal phenotype only	Signal phenotype only	auroc	0.6003	0.6
signal phenotype only	Signal phenotype only	auprc	0.2488	0.2538
signal phenotype only	Signal phenotype only	brier	0.2523	0.2525
signal phenotype only	Signal phenotype only	Calibration intercept	-1.368	
signal phenotype only	Signal phenotype only	Calibration slope	0.4662	
SSL phenotype only	SSL phenotype only	auroc	0.5949	0.5962
SSL phenotype only	SSL phenotype only	auprc	0.246	0.2543
SSL phenotype only	SSL phenotype only	brier	0.246	0.2458
SSL phenotype only	SSL phenotype only	Calibration intercept	-1.38	
SSL phenotype only	SSL phenotype only	Calibration slope	1.456	
signal + signal phenotype	Signal + signal phenotype	auroc	0.6328	0.6321
signal + signal phenotype	Signal + signal phenotype	auprc	0.3495	0.3618
signal + signal phenotype	Signal + signal phenotype	brier	0.2503	0.2506

Continued from Supplementary Table 2, column block 1 of 2.

model	model label	metric	point	bootstrap mean
signal + signal phenotype	Signal + signal phenotype	Calibration intercept	-1.377	
signal + signal phenotype	Signal + signal phenotype	Calibration slope	0.5031	
Signal + SSL phenotype	Signal + SSL phenotype	auroc	0.6667	0.6654
Signal + SSL phenotype	Signal + SSL phenotype	auprc	0.3732	0.3879
Signal + SSL phenotype	Signal + SSL phenotype	brier	0.2483	0.2486
Signal + SSL phenotype	Signal + SSL phenotype	Calibration intercept	-1.426	
Signal + SSL phenotype	Signal + SSL phenotype	Calibration slope	0.5602	
Signal + SSL + phenotype	Signal + SSL + signal phenotype	auroc	0.6168	0.6151
Signal + SSL + phenotype	Signal + SSL + signal phenotype	auprc	0.4016	0.4069
Signal + SSL + phenotype	Signal + SSL + signal phenotype	brier	0.2652	0.2657
Signal + SSL + phenotype	Signal + SSL + signal phenotype	Calibration intercept	-1.359	
Signal + SSL + phenotype	Signal + SSL + signal phenotype	Calibration slope	0.3448	

Table S5: Supplementary Table 2, column block 2 of 2

model	model label	95% CI lower	95% CI upper	Valid bootstrap resamples
Classical signal features	Signal features	0.5254	0.7284	2000
Classical signal features	Signal features	0.2173	0.4941	2000
Classical signal features	Signal features	0.2234	0.2834	2000
Classical signal features	Signal features			
Classical signal features	Signal features			
SSL embeddings	SSL embeddings	0.3795	0.6035	2000
SSL embeddings	SSL embeddings	0.1477	0.3248	2000
SSL embeddings	SSL embeddings	0.2394	0.29	2000
SSL embeddings	SSL embeddings			
SSL embeddings	SSL embeddings			
Signal + SSL	Signal + SSL	0.4927	0.7045	2000
Signal + SSL	Signal + SSL	0.2399	0.5243	2000
Signal + SSL	Signal + SSL	0.2315	0.3066	2000
Signal + SSL	Signal + SSL			
Signal + SSL	Signal + SSL			
signal phenotype only	Signal phenotype only	0.4951	0.697	2000
signal phenotype only	Signal phenotype only	0.168	0.3519	2000
signal phenotype only	Signal phenotype only	0.2298	0.2751	2000
signal phenotype only	Signal phenotype only			
signal phenotype only	Signal phenotype only			
SSL phenotype only	SSL phenotype only	0.4917	0.6912	2000
SSL phenotype only	SSL phenotype only	0.1654	0.3538	2000
SSL phenotype only	SSL phenotype only	0.2368	0.2548	2000
SSL phenotype only	SSL phenotype only			
SSL phenotype only	SSL phenotype only			
signal + signal phenotype	Signal + signal phenotype	0.5243	0.7306	2000
signal + signal phenotype	Signal + signal phenotype	0.221	0.5007	2000
signal + signal phenotype	Signal + signal phenotype	0.2216	0.2807	2000

Continued from Supplementary Table 2, column block 2 of 2.

model	model label	95% CI lower	95% CI upper	Valid bootstrap resamples
signal + signal phenotype	Signal + signal phenotype			
signal + signal phenotype	Signal + signal phenotype			
Signal + SSL phenotype	Signal + SSL phenotype	0.5642	0.7578	2000
Signal + SSL phenotype	Signal + SSL phenotype	0.246	0.5381	2000
Signal + SSL phenotype	Signal + SSL phenotype	0.2189	0.2814	2000
Signal + SSL phenotype	Signal + SSL phenotype			
Signal + SSL phenotype	Signal + SSL phenotype			
Signal + SSL + phenotype	Signal + SSL + signal phenotype	0.5111	0.7191	2000
Signal + SSL + phenotype	Signal + SSL + signal phenotype	0.2605	0.5471	2000
Signal + SSL + phenotype	Signal + SSL + signal phenotype	0.226	0.302	2000
Signal + SSL + phenotype	Signal + SSL + signal phenotype			
Signal + SSL + phenotype	Signal + SSL + signal phenotype			

Supplementary Table 3. Bootstrap pairwise model differences and zero-crossing flags

Complete source table: Supplementary_Table_3_bootstrap_model_differences.csv. Rows: 9; columns: 7. The display is split into source-defined sub-tables and column blocks to preserve a readable table layout; no non-empty source rows or columns are omitted.

Table S6: Supplementary Table 3, column block 1 of 2

comparison	metric	bootstrap mean difference	95% CI lower	95% CI upper
Signal + SSL minus Classical signal features	auroc	-0.0286	-0.1153	0.0541
Signal + SSL minus Classical signal features	auprc	0.031	-0.0527	0.1077
Signal + SSL minus Classical signal features	brier	0.0179	-0.0067	0.0434
Signal + SSL + phenotype minus Classical signal features	auroc	-0.0142	-0.1012	0.0679
Signal + SSL + phenotype minus Classical signal features	auprc	0.0548	-0.0354	0.1519
Signal + SSL + phenotype minus Classical signal features	brier	0.013	-0.0129	0.0386
Signal + SSL minus signal + signal phenotype	auroc	-0.0313	-0.114	0.0509
Signal + SSL minus signal + signal phenotype	auprc	0.0213	-0.0613	0.1056
Signal + SSL minus signal + signal phenotype	brier	0.0201	-0.0046	0.0447

Table S7: Supplementary Table 3, column block 2 of 2

comparison	metric	Valid bootstrap resamples	CI excludes zero
Signal + SSL minus Classical signal features	auroc	2000	False
Signal + SSL minus Classical signal features	auprc	2000	False
Signal + SSL minus Classical signal features	brier	2000	False
Signal + SSL + phenotype minus Classical signal features	auroc	2000	False
Signal + SSL + phenotype minus Classical signal features	auprc	2000	False
Signal + SSL + phenotype minus Classical signal features	brier	2000	False
Signal + SSL minus signal + signal phenotype	auroc	2000	False
Signal + SSL minus signal + signal phenotype	auprc	2000	False
Signal + SSL minus signal + signal phenotype	brier	2000	False

Supplementary Table 4. Signal and SSL phenotype summaries with representative prototype records

Complete source table: Supplementary_Table_4_dynamic_phenotype_summary_and_prototypes.csv. Rows: 8; columns: 39. The display is split into source-defined sub-tables and column blocks to preserve a readable table layout; no non-empty source rows or columns are omitted.

phenotype type: signal phenotype

Table S8: Supplementary Table 4, signal phenotype, column block 1 of 12

phenotype	neonatal risk count	neonatal risk mean	neonatal risk median	cord ph count
1	160	0.1438	0	160
2	157	0.3439	0	157
3	130	0.1923	0	130
4	105	0.1048	0	105

Table S9: Supplementary Table 4, signal phenotype, column block 2 of 12

phenotype	neonatal risk count	cord ph mean	cord ph median	apgar5 count
1	160	7.254	7.27	160
2	157	7.188	7.2	157
3	130	7.236	7.25	130
4	105	7.25	7.25	105

Table S10: Supplementary Table 4, signal phenotype, column block 3 of 12

phenotype	neonatal risk count	apgar5 mean	apgar5 median	FHR mean count
1	160	9.219	9	160
2	157	8.783	9	157
3	130	9.077	9	130
4	105	9.257	9	105

Table S11: Supplementary Table 4, signal phenotype, column block 4 of 12

phenotype	neonatal risk count	FHR mean mean	FHR mean median	FHR sd count
1	160	125.1	125.5	160
2	157	134.6	134.9	157
3	130	148.4	146.3	130
4	105	132.7	133.8	105

Table S12: Supplementary Table 4, signal phenotype, column block 5 of 12

phenotype	neonatal risk count	FHR sd mean	FHR sd median	FHR p05 count
1	160	15.25	15.51	160
2	157	22.22	21.92	157
3	130	14.33	14.26	130
4	105	15.03	15.13	105

Table S13: Supplementary Table 4, signal phenotype, column block 6 of 12

phenotype	neonatal risk count	FHR p05 mean	FHR p05 median	UC mean count
1	160	96.11	96.5	160
2	157	88.23	88.5	157
3	130	122.6	121.6	130
4	105	103.7	104.2	105

Table S14: Supplementary Table 4, signal phenotype, column block 7 of 12

phenotype	neonatal risk count	UC mean mean	UC mean median	UC p95 count
1	160	14.26	14.12	160
2	157	20.22	19.12	157
3	130	16.45	16.08	130
4	105	31.3	29.84	105

Table S15: Supplementary Table 4, signal phenotype, column block 8 of 12

phenotype	neonatal risk count	UC p95 mean	UC p95 median	FHR UC corr count
1	160	40.08	38.5	159
2	157	58.02	56	156
3	130	47.62	46	130
4	105	90.42	92	105

Table S16: Supplementary Table 4, signal phenotype, column block 9 of 12

phenotype	neonatal risk count	FHR UC corr mean	FHR UC corr median	deceleration proxy count count
1	160	0.0105	-0.0061	160
2	157	-0.0139	-0.0237	157
3	130	0.0018	-0.0146	130
4	105	-0.0821	-0.0632	105

Table S17: Supplementary Table 4, signal phenotype, column block 10 of 12

phenotype	neonatal risk count	deceleration proxy count mean	deceleration proxy count median	record id
1	160	1976.3	1850	1122
2	157	3530.5	3404	1016
3	130	1475.1	1406.5	1353
4	105	1894.7	1905	1271

Table S18: Supplementary Table 4, signal phenotype, column block 11 of 12

phenotype	neonatal risk count	distance to centroid	neonatal risk	cord ph
1	160	0.6891	0	7.25
2	157	0.5105	0	7.32
3	130	0.405	0	7.2
4	105	1.001	0	7.31

Table S19: Supplementary Table 4, signal phenotype, column block 12 of 12

phenotype	neonatal risk count	apgar5	n records in phenotype	figure
1	160	10	160	results\figures\phenotype_prototypes\signal-phenotype_1_record_1122.png
2	157	10	157	results\figures\phenotype_prototypes\signal-phenotype_2_record_1016.png
3	130	10	130	results\figures\phenotype_prototypes\signal-phenotype_3_record_1353.png
4	105	10	105	results\figures\phenotype_prototypes\signal-phenotype_4_record_1271.png

phenotype type: SSL phenotype

Table S20: Supplementary Table 4, SSL phenotype, column block 1 of 12

phenotype	neonatal risk count	neonatal risk mean	neonatal risk median	cord ph count
1	125	0.128	0	125
2	139	0.1942	0	139
3	162	0.2284	0	162
4	126	0.2619	0	126

Table S21: Supplementary Table 4, SSL phenotype, column block 2 of 12

phenotype	neonatal risk count	cord ph mean	cord ph median	apgar5 count
1	125	7.259	7.26	125
2	139	7.23	7.24	139
3	162	7.233	7.26	162
4	126	7.197	7.22	126

Table S22: Supplementary Table 4, SSL phenotype, column block 3 of 12

phenotype	neonatal risk count	apgar5 mean	apgar5 median	FHR mean count
1	125	9.184	9	125
2	139	9.194	9	139
3	162	8.988	9	162
4	126	8.921	9	126

Table S23: Supplementary Table 4, SSL phenotype, column block 4 of 12

phenotype	neonatal risk count	FHR mean mean	FHR mean median	FHR sd count
1	125	134.6	134.2	125
2	139	131.6	129.8	139
3	162	136	136.1	162
4	126	136.7	135.3	126

Table S24: Supplementary Table 4, SSL phenotype, column block 5 of 12

phenotype	neonatal risk count	FHR sd mean	FHR sd median	FHR p05 count
1	125	15.51	15.49	125
2	139	16.76	16.64	139
3	162	17.66	17.23	162
4	126	17.77	16.95	126

Table S25: Supplementary Table 4, SSL phenotype, column block 6 of 12

phenotype	neonatal risk count	FHR p05 mean	FHR p05 median	UC mean count
1	125	105.6	105.2	125
2	139	99.78	98.25	139
3	162	99.62	97.12	162
4	126	102	98.25	126

Table S26: Supplementary Table 4, SSL phenotype, column block 7 of 12

phenotype	neonatal risk count	UC mean mean	UC mean median	UC p95 count
1	125	21.88	19.55	125
2	139	19.37	16.93	139
3	162	21.64	19.84	162
4	126	15.45	13.66	126

Table S27: Supplementary Table 4, SSL phenotype, column block 8 of 12

phenotype	neonatal risk count	UC p95 mean	UC p95 median	FHR UC corr count
1	125	68.71	66	125
2	139	52.33	48.5	139
3	162	54.38	53	162
4	126	51.87	47.25	124

Table S28: Supplementary Table 4, SSL phenotype, column block 9 of 12

phenotype	neonatal risk count	FHR UC corr mean	FHR UC corr median	deceleration proxy count count
1	125	-0.0453	-0.0511	125
2	139	0.0239	0.0134	139
3	162	-0.0351	-0.0612	162
4	126	-0.007	-0.013	126

Table S29: Supplementary Table 4, SSL phenotype, column block 10 of 12

phenotype	neonatal risk count	deceleration proxy count mean	deceleration proxy count median	record id
1	125	1932.8	1827	1229
2	139	2036.3	1901	1219
3	162	2772.4	2692.5	2010
4	126	2281.3	2124.5	1344

Table S30: Supplementary Table 4, SSL phenotype, column block 11 of 12

phenotype	neonatal risk count	distance to centroid	neonatal risk	cord ph
1	125	11.99	1	7.13
2	139	18.14	0	7.15
3	162	7.301	0	7.3
4	126	11.2	0	7.24

Table S31: Supplementary Table 4, SSL phenotype, column block 12 of 12

phenotype	neonatal risk count	apgar5	n records in phenotype	figure
1	125	10	167	results\figures\phenotype_prototypes\ssl-phenotype_1_record_1229.png
2	139	9	57	results\figures\phenotype_prototypes\ssl-phenotype_2_record_1219.png
3	162	10	169	results\figures\phenotype_prototypes\ssl-phenotype_3_record_2010.png
4	126	10	159	results\figures\phenotype_prototypes\ssl-phenotype_4_record_1344.png

Supplementary Table 5. Permutation importance by feature family and top individual predictors

Complete source table: Supplementary_Table_5_prediction_explainability.csv. Rows: 48; columns: 16. The display is split into source-defined sub-tables and column blocks to preserve a readable table layout; no non-empty source rows or columns are omitted.

table block: feature family importance

Table S32: Supplementary Table 5, feature family importance, column block 1 of 2

model	feature family	positive importance sum AUPRC	positive importance mean AUPRC	positive importance sum AUROC
Classical signal features	clinical signal	0.1752	0.0146	0.1751
Signal + SSL	clinical signal	0.4874	0.0406	0.2016
Signal + SSL	SSL embedding	2.6	0.0406	0.7787
Signal + SSL + phenotype	clinical signal	0.3688	0.0307	0.1921
Signal + SSL + phenotype	phenotype	0.0235	0.0235	0.0159
Signal + SSL + phenotype	SSL embedding	1.893	0.0296	0.7695

Table S33: Supplementary Table 5, feature family importance, column block 2 of 2

model	feature family	positive importance mean AUROC	n features
Classical signal features	clinical signal	0.0146	12
Signal + SSL	clinical signal	0.0168	12
Signal + SSL	SSL embedding	0.0122	64
Signal + SSL + phenotype	clinical signal	0.016	12
Signal + SSL + phenotype	phenotype	0.0159	1
Signal + SSL + phenotype	SSL embedding	0.012	64

table block: top feature importance

Table S34: Supplementary Table 5, top feature importance, column block 1 of 3

model	feature family	feature	importance mean AUPRC	importance std AUPRC
Classical signal features	clinical signal	FHR mean	0.0579	0.0329
Classical signal features	clinical signal	FHR missing fraction	0.0466	0.0258
Classical signal features	clinical signal	deceleration proxy count	0.0328	0.0491
Classical signal features	clinical signal	FHR p05	0.019	0.0199
Classical signal features	clinical signal	FHR UC corr	0.0189	0.0232
Classical signal features	clinical signal	UC missing fraction	0	0
Classical signal features	clinical signal	UC sd	-9.85e-06	0.0002
Classical signal features	clinical signal	FHR sd	-0.0007	0.0029
Classical signal features	clinical signal	UC mean	-0.0017	0.0078
Classical signal features	clinical signal	FHR p95	-0.0019	0.0024
Classical signal features	clinical signal	UC p95	-0.0061	0.0078
Classical signal features	clinical signal	FHR p50	-0.0275	0.031
Signal + SSL	clinical signal	FHR missing fraction	0.1642	0.0343
Signal + SSL	SSL embedding	emb 55	0.1584	0.0346
Signal + SSL	SSL embedding	emb 32	0.1387	0.0273
Signal + SSL	SSL embedding	emb 17	0.1179	0.0318
Signal + SSL	SSL embedding	emb 19	0.1178	0.0266
Signal + SSL	SSL embedding	emb 43	0.1004	0.0317
Signal + SSL	clinical signal	FHR mean	0.0989	0.0314
Signal + SSL	SSL embedding	emb 18	0.0886	0.0271
Signal + SSL	SSL embedding	emb 63	0.0818	0.0321
Signal + SSL	SSL embedding	emb 52	0.0783	0.0193
Signal + SSL	SSL embedding	emb 26	0.0775	0.0203
Signal + SSL	SSL embedding	emb 25	0.0769	0.0223
Signal + SSL	SSL embedding	emb 42	0.0709	0.0247
Signal + SSL	SSL embedding	emb 44	0.0699	0.0323
Signal + SSL	SSL embedding	emb 29	0.068	0.0248

Continued from Supplementary Table 5, top feature importance, column block 1 of 3.

model	feature family	feature	importance mean AUPRC	importance std AUPRC
Signal + SSL + phenotype	clinical signal	FHR missing fraction	0.1508	0.0354
Signal + SSL + phenotype	SSL embedding	emb 55	0.1404	0.037
Signal + SSL + phenotype	SSL embedding	emb 32	0.1263	0.0276
Signal + SSL + phenotype	SSL embedding	emb 17	0.1102	0.0307
Signal + SSL + phenotype	SSL embedding	emb 19	0.0895	0.0259
Signal + SSL + phenotype	SSL embedding	emb 43	0.0803	0.0308
Signal + SSL + phenotype	clinical signal	FHR mean	0.0801	0.0304
Signal + SSL + phenotype	SSL embedding	emb 63	0.0683	0.0323
Signal + SSL + phenotype	SSL embedding	emb 52	0.0673	0.0227
Signal + SSL + phenotype	SSL embedding	emb 26	0.0627	0.0206
Signal + SSL + phenotype	SSL embedding	emb 25	0.062	0.0234
Signal + SSL + phenotype	SSL embedding	emb 18	0.0613	0.0291
Signal + SSL + phenotype	SSL embedding	emb 42	0.0554	0.0234
Signal + SSL + phenotype	SSL embedding	emb 53	0.0534	0.0223
Signal + SSL + phenotype	SSL embedding	emb 44	0.0527	0.0304

Table S35: Supplementary Table 5, top feature importance, column block 2 of 3

model	feature family	importance mean AUROC	importance std AUROC	feature label
Classical signal features	clinical signal	0.0736	0.0204	Mean FHR
Classical signal features	clinical signal	0.0344	0.0288	FHR missing fraction
Classical signal features	clinical signal	0.0508	0.0365	Deceleration-proxy burden
Classical signal features	clinical signal	0.0144	0.0138	FHR 5th percentile
Classical signal features	clinical signal	-0.0044	0.0182	FHR-UC correlation
Classical signal features	clinical signal	0	0	UC missing fraction
Classical signal features	clinical signal	0.0001	0.0002	UC variability (SD)
Classical signal features	clinical signal	0.0017	0.0015	FHR variability (SD)
Classical signal features	clinical signal	-0.0002	0.0058	Mean uterine contraction
Classical signal features	clinical signal	2.97e-05	0.0003	FHR 95th percentile
Classical signal features	clinical signal	-0.0011	0.0059	UC 95th percentile
Classical signal features	clinical signal	-0.0103	0.0168	Median FHR
Signal + SSL	clinical signal	0.0846	0.0327	FHR missing fraction
Signal + SSL	SSL embedding	0.0702	0.0246	SSL dimension 55
Signal + SSL	SSL embedding	0.0503	0.0224	SSL dimension 32
Signal + SSL	SSL embedding	0.0187	0.0239	SSL dimension 17
Signal + SSL	SSL embedding	0.0119	0.0316	SSL dimension 19
Signal + SSL	SSL embedding	0.0335	0.0221	SSL dimension 43
Signal + SSL	clinical signal	0.0508	0.0162	Mean FHR
Signal + SSL	SSL embedding	0.025	0.0174	SSL dimension 18
Signal + SSL	SSL embedding	0.0358	0.0306	SSL dimension 63
Signal + SSL	SSL embedding	0.0112	0.0185	SSL dimension 52
Signal + SSL	SSL embedding	0.0128	0.0194	SSL dimension 26
Signal + SSL	SSL embedding	0.0647	0.0237	SSL dimension 25
Signal + SSL	SSL embedding	-0.0051	0.0204	SSL dimension 42
Signal + SSL	SSL embedding	-0.0278	0.021	SSL dimension 44
Signal + SSL	SSL embedding	0.014	0.0151	SSL dimension 29

Continued from Supplementary Table 5, top feature importance, column block 2 of 3.

model	feature family	importance mean AUROC	importance std AUROC	feature label
Signal + SSL + phenotype	clinical signal	0.0868	0.0327	FHR missing fraction
Signal + SSL + phenotype	SSL embedding	0.0721	0.0241	SSL dimension 55
Signal + SSL + phenotype	SSL embedding	0.0523	0.0224	SSL dimension 32
Signal + SSL + phenotype	SSL embedding	0.0214	0.0233	SSL dimension 17
Signal + SSL + phenotype	SSL embedding	0.009	0.0295	SSL dimension 19
Signal + SSL + phenotype	SSL embedding	0.0332	0.0214	SSL dimension 43
Signal + SSL + phenotype	clinical signal	0.05	0.0156	Mean FHR
Signal + SSL + phenotype	SSL embedding	0.0372	0.031	SSL dimension 63
Signal + SSL + phenotype	SSL embedding	0.0137	0.0189	SSL dimension 52
Signal + SSL + phenotype	SSL embedding	0.0117	0.0188	SSL dimension 26
Signal + SSL + phenotype	SSL embedding	0.0638	0.0243	SSL dimension 25
Signal + SSL + phenotype	SSL embedding	0.023	0.0156	SSL dimension 18
Signal + SSL + phenotype	SSL embedding	-0.0049	0.0197	SSL dimension 42
Signal + SSL + phenotype	SSL embedding	0.0538	0.0192	SSL dimension 53
Signal + SSL + phenotype	SSL embedding	-0.0268	0.0202	SSL dimension 44

Table S36: Supplementary Table 5, top feature importance, column block 3 of 3

model	feature family	rank AUPRC	rank AUROC
Classical signal features	clinical signal	1	1
Classical signal features	clinical signal	2	3
Classical signal features	clinical signal	3	2
Classical signal features	clinical signal	4	4
Classical signal features	clinical signal	5	11
Classical signal features	clinical signal	6	8
Classical signal features	clinical signal	7	6
Classical signal features	clinical signal	8	5
Classical signal features	clinical signal	9	9
Classical signal features	clinical signal	10	7
Classical signal features	clinical signal	11	10
Classical signal features	clinical signal	12	12
Signal + SSL	clinical signal	1	1
Signal + SSL	SSL embedding	2	2
Signal + SSL	SSL embedding	3	6
Signal + SSL	SSL embedding	4	18
Signal + SSL	SSL embedding	5	31
Signal + SSL	SSL embedding	6	10
Signal + SSL	clinical signal	7	5
Signal + SSL	SSL embedding	8	13
Signal + SSL	SSL embedding	9	8
Signal + SSL	SSL embedding	10	32
Signal + SSL	SSL embedding	11	29
Signal + SSL	SSL embedding	12	3
Signal + SSL	SSL embedding	13	58
Signal + SSL	SSL embedding	14	76
Signal + SSL	SSL embedding	15	26

Continued from Supplementary Table 5, top feature importance, column block 3 of 3.

model	feature family	rank AUPRC	rank AUROC
Signal + SSL + phenotype	clinical signal	1	1
Signal + SSL + phenotype	SSL embedding	2	2
Signal + SSL + phenotype	SSL embedding	3	5
Signal + SSL + phenotype	SSL embedding	4	17
Signal + SSL + phenotype	SSL embedding	5	35
Signal + SSL + phenotype	SSL embedding	6	9
Signal + SSL + phenotype	clinical signal	7	6
Signal + SSL + phenotype	SSL embedding	8	8
Signal + SSL + phenotype	SSL embedding	9	23
Signal + SSL + phenotype	SSL embedding	10	29
Signal + SSL + phenotype	SSL embedding	11	3
Signal + SSL + phenotype	SSL embedding	12	15
Signal + SSL + phenotype	SSL embedding	13	58
Signal + SSL + phenotype	SSL embedding	14	4
Signal + SSL + phenotype	SSL embedding	15	77

Supplementary Table 6. Performance and model-difference sensitivity after removing missingness features

Complete source table: Supplementary_Table_6_missingness_sensitivity.csv. Rows: 39; columns: 12. The display is split into source-defined sub-tables and column blocks to preserve a readable table layout; no non-empty source rows or columns are omitted.

table block: scenario metrics

Table S37: Supplementary Table 6, scenario metrics, column block 1 of 2

scenario	model	metric	point	bootstrap mean
full features	Classical signal features	auroc	0.6303	0.6293
full features	Classical signal features	auprc	0.3366	0.3522
full features	Classical signal features	brier	0.2525	0.2528
full features	Signal + SSL	auroc	0.6025	0.6007
full features	Signal + SSL	auprc	0.3765	0.3832
full features	Signal + SSL	brier	0.2701	0.2707
full features	Signal + SSL + phenotype	auroc	0.6168	0.6151
full features	Signal + SSL + phenotype	auprc	0.4016	0.4069
full features	Signal + SSL + phenotype	brier	0.2652	0.2657
no missingness features	Classical signal features	auroc	0.6428	0.642
no missingness features	Classical signal features	auprc	0.3765	0.39
no missingness features	Classical signal features	brier	0.2482	0.2482
no missingness features	Signal + SSL	auroc	0.5887	0.5872
no missingness features	Signal + SSL	auprc	0.3951	0.3998
no missingness features	Signal + SSL	brier	0.2709	0.2713
no missingness features	Signal + SSL + phenotype	auroc	0.5945	0.593
no missingness features	Signal + SSL + phenotype	auprc	0.3982	0.4032
no missingness features	Signal + SSL + phenotype	brier	0.2672	0.2677

Table S38: Supplementary Table 6, scenario metrics, column block 2 of 2

scenario	model	95% CI lower	95% CI upper	Valid bootstrap resamples
full features	Classical signal features	0.5254	0.7284	2000
full features	Classical signal features	0.2173	0.4941	2000
full features	Classical signal features	0.2234	0.2834	2000
full features	Signal + SSL	0.4927	0.7045	2000
full features	Signal + SSL	0.2399	0.5243	2000
full features	Signal + SSL	0.2315	0.3066	2000
full features	Signal + SSL + phenotype	0.5111	0.7191	2000
full features	Signal + SSL + phenotype	0.2605	0.5471	2000
full features	Signal + SSL + phenotype	0.226	0.302	2000
no missingness features	Classical signal features	0.5348	0.7461	2000
no missingness features	Classical signal features	0.2418	0.5493	2000
no missingness features	Classical signal features	0.2216	0.2754	2000
no missingness features	Signal + SSL	0.4729	0.7018	2000
no missingness features	Signal + SSL	0.2507	0.547	2000
no missingness features	Signal + SSL	0.2352	0.3064	2000
no missingness features	Signal + SSL + phenotype	0.4822	0.7077	2000
no missingness features	Signal + SSL + phenotype	0.2537	0.5486	2000
no missingness features	Signal + SSL + phenotype	0.2315	0.3025	2000

table block: scenario differences

Table S39: Supplementary Table 6, scenario differences, column block 1 of 2

scenario	metric	95% CI lower	95% CI upper	Valid bootstrap resamples
full features	auroc	-0.1153	0.0541	2000
full features	auprc	-0.0527	0.1077	2000
full features	brier	-0.0067	0.0434	2000
full features	auroc	-0.1012	0.0679	2000
full features	auprc	-0.0354	0.1519	2000
full features	brier	-0.0129	0.0386	2000
no missingness features	auroc	-0.1207	0.0085	2000
no missingness features	auprc	-0.0702	0.087	2000
no missingness features	brier	0.0026	0.0443	2000
no missingness features	auroc	-0.12	0.0164	2000
no missingness features	auprc	-0.0691	0.0894	2000
no missingness features	brier	-0.0011	0.0408	2000
no missingness features vs full features	auroc	-0.042	0.0628	2000
no missingness features vs full features	auprc	-0.038	0.1149	2000
no missingness features vs full features	brier	-0.0194	0.0105	2000
no missingness features vs full features	auroc	-0.0707	0.0441	2000
no missingness features vs full features	auprc	-0.0313	0.068	2000
no missingness features vs full features	brier	-0.0192	0.0214	2000
no missingness features vs full features	auroc	-0.0818	0.037	2000
no missingness features vs full features	auprc	-0.0526	0.0415	2000

Continued from Supplementary Table 6, scenario differences, column block 1 of 2.

scenario	metric	95% CI lower	95% CI upper	Valid bootstrap resamples
no missingness features vs full features	brier	-0.0185	0.0229	2000

Table S40: Supplementary Table 6, scenario differences, column block 2 of 2

scenario	metric	comparison type	comparison	bootstrap mean difference
full features	auroc	model difference within scenario	Signal + SSL minus Classical signal features	-0.0286
full features	auprc	model difference within scenario	Signal + SSL minus Classical signal features	0.031
full features	brier	model difference within scenario	Signal + SSL minus Classical signal features	0.0179
full features	auroc	model difference within scenario	Signal + SSL + phenotype minus Classical signal features	-0.0142
full features	auprc	model difference within scenario	Signal + SSL + phenotype minus Classical signal features	0.0548
full features	brier	model difference within scenario	Signal + SSL + phenotype minus Classical signal features	0.013
no missingness features	auroc	model difference within scenario	Signal + SSL minus Classical signal features	-0.0549
no missingness features	auprc	model difference within scenario	Signal + SSL minus Classical signal features	0.0098
no missingness features	brier	model difference within scenario	Signal + SSL minus Classical signal features	0.0231
no missingness features	auroc	model difference within scenario	Signal + SSL + phenotype minus Classical signal features	-0.0491
no missingness features	auprc	model difference within scenario	Signal + SSL + phenotype minus Classical signal features	0.0132
no missingness features	brier	model difference within scenario	Signal + SSL + phenotype minus Classical signal features	0.0194

Continued from Supplementary Table 6, scenario differences, column block 2 of 2.

scenario	metric	comparison type	comparison	bootstrap mean difference
no missingness features vs full features	auroc	no missingness minus full	Classical signal features	0.0127
no missingness features vs full features	auprc	no missingness minus full	Classical signal features	0.0378
no missingness features vs full features	brier	no missingness minus full	Classical signal features	-0.0046
no missingness features vs full features	auroc	no missingness minus full	Signal + SSL	-0.0136
no missingness features vs full features	auprc	no missingness minus full	Signal + SSL	0.0166
no missingness features vs full features	brier	no missingness minus full	Signal + SSL	0.0006
no missingness features vs full features	auroc	no missingness minus full	Signal + SSL + phenotype	-0.0221
no missingness features vs full features	auprc	no missingness minus full	Signal + SSL + phenotype	-0.0037
no missingness features vs full features	brier	no missingness minus full	Signal + SSL + phenotype	0.0019

Supplementary Table 7. Cross-fitted recalibration metrics, recalibration differences, and decision curve summaries

Complete source table: Supplementary_Table_7_recalibration_and_decision_curve.csv. Rows: 126; columns: 15. The display is split into source-defined sub-tables and column blocks to preserve a readable table layout; no non-empty source rows or columns are omitted.

table block: Recalibration metrics

Table S41: Supplementary Table 7, Recalibration metrics, column block 1 of 2

model	calibration method	metric	point	bootstrap mean
Classical signal features	raw	auroc	0.6303	0.6293
Classical signal features	raw	auprc	0.3366	0.3522
Classical signal features	raw	brier	0.2525	0.2528
Classical signal features	raw	Calibration intercept	-1.384	-1.4
Classical signal features	raw	Calibration slope	0.4876	0.4923
Classical signal features	raw	Expected calibration error	0.2745	0.2745
Classical signal features	raw	Mean prediction error	0.2745	0.2735
Classical signal features	platt	auroc	0.6303	0.6293
Classical signal features	platt	auprc	0.3366	0.3522
Classical signal features	platt	brier	0.1585	0.1591
Classical signal features	platt	Calibration intercept	-0.4845	-0.4923
Classical signal features	platt	Calibration slope	0.6906	0.6972
Classical signal features	platt	Expected calibration error	0.0527	0.074
Classical signal features	platt	Mean prediction error	0.0212	0.0203
Classical signal features	isotonic	auroc	0.639	0.6382
Classical signal features	isotonic	auprc	0.3064	0.3175
Classical signal features	isotonic	brier	0.1659	0.1664
Classical signal features	isotonic	Calibration intercept	-0.716	-0.7232
Classical signal features	isotonic	Calibration slope	0.4983	0.5031
Classical signal features	isotonic	Expected calibration error	0.0625	0.0845
Classical signal features	isotonic	Mean prediction error	0.0237	0.0228
Signal + SSL	raw	auroc	0.6025	0.6007
Signal + SSL	raw	auprc	0.3765	0.3832
Signal + SSL	raw	brier	0.2701	0.2707
Signal + SSL	raw	Calibration intercept	-1.36	-1.377
Signal + SSL	raw	Calibration slope	0.3136	0.3132
Signal + SSL	raw	Expected calibration error	0.2601	0.2754

Continued from Supplementary Table 7, Recalibration metrics, column block 1 of 2.

model	calibration method	metric	point	bootstrap mean
Signal + SSL	raw	Mean prediction error	0.2601	0.2593
Signal + SSL	platt	auroc	0.6025	0.6007
Signal + SSL	platt	auprc	0.3765	0.3832
Signal + SSL	platt	brier	0.1576	0.1583
Signal + SSL	platt	Calibration intercept	-0.1999	-0.2188
Signal + SSL	platt	Calibration slope	0.947	0.9458
Signal + SSL	platt	Expected calibration error	0.0491	0.0701
Signal + SSL	platt	Mean prediction error	0.0226	0.0216
Signal + SSL	isotonic	auroc	0.5916	0.5902
Signal + SSL	isotonic	auprc	0.254	0.2607
Signal + SSL	isotonic	brier	0.1658	0.1665
Signal + SSL	isotonic	Calibration intercept	-0.8159	-0.8166
Signal + SSL	isotonic	Calibration slope	0.4215	0.433
Signal + SSL	isotonic	Expected calibration error	0.0644	0.0772
Signal + SSL	isotonic	Mean prediction error	0.0236	0.0227
Signal + SSL + phenotype	raw	auroc	0.6168	0.6151
Signal + SSL + phenotype	raw	auprc	0.4016	0.4069
Signal + SSL + phenotype	raw	brier	0.2652	0.2657
Signal + SSL + phenotype	raw	Calibration intercept	-1.359	-1.376
Signal + SSL + phenotype	raw	Calibration slope	0.3448	0.3461
Signal + SSL + phenotype	raw	Expected calibration error	0.2568	0.2699
Signal + SSL + phenotype	raw	Mean prediction error	0.2568	0.2562

Continued from Supplementary Table 7, Recalibration metrics, column block 1 of 2.

model	calibration method	metric	point	bootstrap mean
Signal + SSL + phenotype	platt	auroc	0.6168	0.6151
Signal + SSL + phenotype	platt	auprc	0.4016	0.4069
Signal + SSL + phenotype	platt	brier	0.1566	0.1572
Signal + SSL + phenotype	platt	Calibration intercept	0.077	0.0641
Signal + SSL + phenotype	platt	Calibration slope	1.165	1.169
Signal + SSL + phenotype	platt	Expected calibration error	0.0322	0.0648
Signal + SSL + phenotype	platt	Mean prediction error	0.0192	0.0183
Signal + SSL + phenotype	isotonic	auroc	0.5896	0.5888
Signal + SSL + phenotype	isotonic	auprc	0.2672	0.2718
Signal + SSL + phenotype	isotonic	brier	0.1623	0.163
Signal + SSL + phenotype	isotonic	Calibration intercept	-0.6669	-0.6662
Signal + SSL + phenotype	isotonic	Calibration slope	0.5354	0.5495
Signal + SSL + phenotype	isotonic	Expected calibration error	0.0544	0.0634
Signal + SSL + phenotype	isotonic	Mean prediction error	0.0198	0.019

Table S42: Supplementary Table 7, Recalibration metrics, column block 2 of 2

model	calibration method	95% CI lower	95% CI upper	Valid bootstrap resamples
Classical signal features	raw	0.5254	0.7284	2000
Classical signal features	raw	0.2173	0.4941	2000
Classical signal features	raw	0.2234	0.2834	2000
Classical signal features	raw	-1.829	-1.014	2000
Classical signal features	raw	0.1338	0.875	2000
Classical signal features	raw	0.2102	0.3376	2000
Classical signal features	raw	0.2086	0.3371	2000
Classical signal features	platt	0.5254	0.7284	2000
Classical signal features	platt	0.2173	0.4941	2000
Classical signal features	platt	0.1266	0.1945	2000
Classical signal features	platt	-1.289	0.1958	2000
Classical signal features	platt	0.1896	1.239	2000
Classical signal features	platt	0.0341	0.1218	2000
Classical signal features	platt	-0.0427	0.0792	2000
Classical signal features	isotonic	0.5357	0.7316	2000
Classical signal features	isotonic	0.2023	0.4411	2000
Classical signal features	isotonic	0.1301	0.2044	2000
Classical signal features	isotonic	-1.395	-0.1362	2000
Classical signal features	isotonic	0.1281	0.9124	2000
Classical signal features	isotonic	0.0378	0.1342	2000
Classical signal features	isotonic	-0.0419	0.0849	2000
Signal + SSL	raw	0.4927	0.7045	2000
Signal + SSL	raw	0.2399	0.5243	2000
Signal + SSL	raw	0.2315	0.3066	2000
Signal + SSL	raw	-1.806	-0.9946	2000
Signal + SSL	raw	0.0476	0.5935	2000
Signal + SSL	raw	0.2175	0.3365	2000

Continued from Supplementary Table 7, Recalibration metrics, column block 2 of 2.

model	calibration method	95% CI lower	95% CI upper	Valid bootstrap resamples
Signal + SSL	raw	0.1892	0.3275	2000
Signal + SSL	platt	0.4927	0.7045	2000
Signal + SSL	platt	0.2399	0.5243	2000
Signal + SSL	platt	0.1265	0.1923	2000
Signal + SSL	platt	-1.242	0.7797	2000
Signal + SSL	platt	0.144	1.793	2000
Signal + SSL	platt	0.0298	0.1184	2000
Signal + SSL	platt	-0.0419	0.0803	2000
Signal + SSL	isotonic	0.4884	0.6902	2000
Signal + SSL	isotonic	0.17	0.3664	2000
Signal + SSL	isotonic	0.1332	0.2028	2000
Signal + SSL	isotonic	-1.577	-0.0402	2000
Signal + SSL	isotonic	-0.0321	1.026	2000
Signal + SSL	isotonic	0.027	0.1366	2000
Signal + SSL	isotonic	-0.0418	0.0828	2000
Signal + SSL + phenotype	raw	0.5111	0.7191	2000
Signal + SSL + phenotype	raw	0.2605	0.5471	2000
Signal + SSL + phenotype	raw	0.226	0.302	2000
Signal + SSL + phenotype	raw	-1.809	-0.9874	2000
Signal + SSL + phenotype	raw	0.0861	0.6295	2000
Signal + SSL + phenotype	raw	0.2083	0.3374	2000
Signal + SSL + phenotype	raw	0.1869	0.3227	2000

Continued from Supplementary Table 7, Recalibration metrics, column block 2 of 2.

model	calibration method	95% CI lower	95% CI upper	Valid bootstrap resamples
Signal + SSL + phenotype	platt	0.5111	0.7191	2000
Signal + SSL + phenotype	platt	0.2605	0.5471	2000
Signal + SSL + phenotype	platt	0.1254	0.1916	2000
Signal + SSL + phenotype	platt	-1.04	1.144	2000
Signal + SSL + phenotype	platt	0.2914	2.125	2000
Signal + SSL + phenotype	platt	0.0284	0.1098	2000
Signal + SSL + phenotype	platt	-0.0446	0.0769	2000
Signal + SSL + phenotype	isotonic	0.4914	0.6892	2000
Signal + SSL + phenotype	isotonic	0.1762	0.3749	2000
Signal + SSL + phenotype	isotonic	0.1293	0.199	2000
Signal + SSL + phenotype	isotonic	-1.459	0.161	2000
Signal + SSL + phenotype	isotonic	0.0309	1.193	2000
Signal + SSL + phenotype	isotonic	0.0199	0.113	2000
Signal + SSL + phenotype	isotonic	-0.0453	0.0784	2000

table block: recalibration differences

Table S43: Supplementary Table 7, recalibration differences, column block 1 of 2

model	metric	95% CI lower	95% CI upper	Valid bootstrap resamples
Classical signal features	brier	-0.1318	-0.0545	2000
Classical signal features	Calibration slope	0.0557	0.3639	2000
Classical signal features	Expected calibration error	-0.2573	-0.1232	2000
Classical signal features	auprc	0	0	2000
Classical signal features	auroc	0	0	2000
Classical signal features	brier	-0.1228	-0.0474	2000
Classical signal features	Calibration slope	-0.1546	0.1817	2000
Classical signal features	Expected calibration error	-0.2543	-0.1042	2000
Classical signal features	auprc	-0.0801	-0.0003	2000
Classical signal features	auroc	-0.0107	0.0311	2000
Signal + SSL	brier	-0.1551	-0.0689	2000
Signal + SSL	Calibration slope	0.0965	1.199	2000
Signal + SSL	Expected calibration error	-0.2551	-0.1422	2000
Signal + SSL	auprc	0	0	2000
Signal + SSL	auroc	0	0	2000
Signal + SSL	brier	-0.1447	-0.0631	2000
Signal + SSL	Calibration slope	-0.1522	0.4647	2000
Signal + SSL	Expected calibration error	-0.256	-0.131	2000
Signal + SSL	auprc	-0.2113	-0.0423	2000
Signal + SSL	auroc	-0.0351	0.0141	2000
Signal + SSL + phenotype	brier	-0.1516	-0.0643	2000
Signal + SSL + phenotype	Calibration slope	0.2053	1.496	2000
Signal + SSL + phenotype	Expected calibration error	-0.2588	-0.1294	2000
Signal + SSL + phenotype	auprc	0	0	2000

Continued from Supplementary Table 7, recalibration differences, column block 1 of 2.

model	metric	95% CI lower	95% CI upper	Valid bootstrap resamples
Signal + SSL + phenotype	auroc	0	0	2000
Signal + SSL + phenotype	brier	-0.1434	-0.0608	2000
Signal + SSL + phenotype	Calibration slope	-0.0976	0.6107	2000
Signal + SSL + phenotype	Expected calibration error	-0.263	-0.1326	2000
Signal + SSL + phenotype	auprc	-0.2366	-0.0471	2000
Signal + SSL + phenotype	auroc	-0.0606	0.01	2000

Table S44: Supplementary Table 7, recalibration differences, column block 2 of 2

model	metric	comparison	bootstrap mean difference
Classical signal features	brier	platt minus raw	-0.0937
Classical signal features	Calibration slope	platt minus raw	0.2049
Classical signal features	Expected calibration error	platt minus raw	-0.2005
Classical signal features	auprc	platt minus raw	0
Classical signal features	auroc	platt minus raw	0
Classical signal features	brier	isotonic minus raw	-0.0864
Classical signal features	Calibration slope	isotonic minus raw	0.0108
Classical signal features	Expected calibration error	isotonic minus raw	-0.19
Classical signal features	auprc	isotonic minus raw	-0.0346
Classical signal features	auroc	isotonic minus raw	0.0089
Signal + SSL	brier	platt minus raw	-0.1124
Signal + SSL	Calibration slope	platt minus raw	0.6327
Signal + SSL	Expected calibration error	platt minus raw	-0.2053
Signal + SSL	auprc	platt minus raw	0
Signal + SSL	auroc	platt minus raw	0
Signal + SSL	brier	isotonic minus raw	-0.1042
Signal + SSL	Calibration slope	isotonic minus raw	0.1199
Signal + SSL	Expected calibration error	isotonic minus raw	-0.1982
Signal + SSL	auprc	isotonic minus raw	-0.1225
Signal + SSL	auroc	isotonic minus raw	-0.0105
Signal + SSL + phenotype	brier	platt minus raw	-0.1085
Signal + SSL + phenotype	Calibration slope	platt minus raw	0.8225
Signal + SSL + phenotype	Expected calibration error	platt minus raw	-0.205
Signal + SSL + phenotype	auprc	platt minus raw	0
Signal + SSL + phenotype	auroc	platt minus raw	0
Signal + SSL + phenotype	brier	isotonic minus raw	-0.1027
Signal + SSL + phenotype	Calibration slope	isotonic minus raw	0.2035

Continued from Supplementary Table 7, recalibration differences, column block 2 of 2.

model	metric	comparison	bootstrap mean difference
Signal + SSL + phenotype	Expected calibration error	isotonic minus raw	-0.2065
Signal + SSL + phenotype	auprc	isotonic minus raw	-0.1351
Signal + SSL + phenotype	auroc	isotonic minus raw	-0.0263

table block: Decision curve summary

Table S45: Supplementary Table 7, Decision curve summary

threshold range	strategy	mean net benefit	Mean delta vs raw final model
0.05-0.20	Classical signal features Isotonic	0.0955	-0.0056
0.05-0.20	Classical signal features Platt	0.0992	-0.002
0.05-0.20	Classical signal features Raw	0.0944	-0.0068
0.05-0.20	Signal + SSL Isotonic	0.0885	-0.0127
0.05-0.20	Signal + SSL Platt	0.0939	-0.0073
0.05-0.20	Signal + SSL + phenotype Isotonic	0.0914	-0.0098
0.05-0.20	Signal + SSL + phenotype Platt	0.0961	-0.0051
0.05-0.20	Signal + SSL + phenotype Raw	0.1012	
0.05-0.20	Signal + SSL Raw	0.097	-0.0042
0.05-0.20	Treat all	0.0887	-0.0125
0.05-0.20	Treat none	0	-0.1012
0.10-0.30	Classical signal features Isotonic	0.0483	0.0223
0.10-0.30	Classical signal features Platt	0.045	0.019
0.10-0.30	Classical signal features Raw	0.0262	0.0001
0.10-0.30	Signal + SSL Isotonic	0.0376	0.0115
0.10-0.30	Signal + SSL Platt	0.0405	0.0144
0.10-0.30	Signal + SSL + phenotype Isotonic	0.0389	0.0128
0.10-0.30	Signal + SSL + phenotype Platt	0.0442	0.0181
0.10-0.30	Signal + SSL + phenotype Raw	0.0261	
0.10-0.30	Signal + SSL Raw	0.0263	0.0002
0.10-0.30	Treat all	0.0003	-0.0258
0.10-0.30	Treat none	0	-0.0261
0.05-0.50	Classical signal features Isotonic	0.0326	0.0671
0.05-0.50	Classical signal features Platt	0.0362	0.0708
0.05-0.50	Classical signal features Raw	-0.0397	-0.0051
0.05-0.50	Signal + SSL Isotonic	0.0254	0.0599

Continued from Supplementary Table 7, Decision curve summary.

threshold range	strategy	mean net benefit	Mean delta vs raw final model
0.05-0.50	Signal + SSL Platt	0.0389	0.0734
0.05-0.50	Signal + SSL + phenotype Isotonic	0.0311	0.0656
0.05-0.50	Signal + SSL + phenotype Platt	0.0405	0.075
0.05-0.50	Signal + SSL + phenotype Raw	-0.0345	
0.05-0.50	Signal + SSL Raw	-0.0385	-0.004
0.05-0.50	Treat all	-0.136	-0.1015
0.05-0.50	Treat none	0	0.0345

Supplementary Table 8. CTGDL-FHRMA domain-shift and phenotype-assignment assessment

Complete source table: Supplementary_Table_8_external_representation_validation.csv. Rows: 12; columns: 5. The display is split into source-defined sub-tables and column blocks to preserve a readable table layout; no non-empty source rows or columns are omitted.

table block: external embedding metrics

Table S46: Supplementary Table 8, external embedding metrics

metric	value
ctu records	552
ctgdl fhrma records	135
domain classifier AUROC ctgdl vs ctu	0.9088
PCA explained variance pc1 pc2	0.5092
ctgdl assigned SSL phenotype 1	56
ctgdl assigned SSL phenotype 2	1
ctgdl assigned SSL phenotype 3	62
ctgdl assigned SSL phenotype 4	16

table block: assigned SSL phenotype counts

Table S47: Supplementary Table 8, assigned SSL phenotype counts

assigned ctu SSL phenotype	n records
1	56
2	1
3	62
4	16

Supplementary Table 9. Reviewer-requested sensitivity analyses for development-only model selection, SSL PCA dimensionality reduction, elastic-net regularization and pH-threshold outcome definitions

Complete source table: Supplementary_Table_9_reviewer_sensitivity_analyses.csv. Rows: 24; columns: 24. The display is split into source-defined sub-tables and column blocks to preserve a readable table layout; no non-empty source rows or columns are omitted.

analysis type: Development model selection

Table S48: Supplementary Table 9, Development model selection, column block 1 of 3

config	model	selection split	Development train records	Development validation records
p80 d64 l3 channel cw02	Signal + SSL	75% original-training records fit / 25% original-training records validation	289	97
p80 d64 l3 channel cw02	Signal + SSL + phenotype	75% original-training records fit / 25% original-training records validation	289	97
p60 d64 l2 mixed cw02	Signal + SSL	75% original-training records fit / 25% original-training records validation	289	97
p60 d64 l2 mixed cw02	Signal + SSL + phenotype	75% original-training records fit / 25% original-training records validation	289	97
p80 d48 l2 mixed cw01	Signal + SSL	75% original-training records fit / 25% original-training records validation	289	97
p80 d48 l2 mixed cw01	Signal + SSL + phenotype	75% original-training records fit / 25% original-training records validation	289	97
p120 d64 l2 block cw02	Signal + SSL	75% original-training records fit / 25% original-training records validation	289	97
p120 d64 l2 block cw02	Signal + SSL + phenotype	75% original-training records fit / 25% original-training records validation	289	97

Table S49: Supplementary Table 9, Development model selection, column block 2 of 3

config	model	Development train events	Development validation events	AUROC
p80 d64 l3 channel cw02	Signal + SSL	59	20	0.7078
p80 d64 l3 channel cw02	Signal + SSL + phenotype	59	20	0.7045
p60 d64 l2 mixed cw02	Signal + SSL	59	20	0.6942
p60 d64 l2 mixed cw02	Signal + SSL + phenotype	59	20	0.6981
p80 d48 l2 mixed cw01	Signal + SSL	59	20	0.7396
p80 d48 l2 mixed cw01	Signal + SSL + phenotype	59	20	0.75
p120 d64 l2 block cw02	Signal + SSL	59	20	0.6519
p120 d64 l2 block cw02	Signal + SSL + phenotype	59	20	0.6597

Table S50: Supplementary Table 9, Development model selection, column block 3 of 3

config	model	AUPRC	Brier score	event prevalence
p80 d64 l3 channel cw02	Signal + SSL	0.3569	0.2411	0.2062
p80 d64 l3 channel cw02	Signal + SSL + phenotype	0.3569	0.2414	0.2062
p60 d64 l2 mixed cw02	Signal + SSL	0.3415	0.2282	0.2062
p60 d64 l2 mixed cw02	Signal + SSL + phenotype	0.3564	0.2287	0.2062
p80 d48 l2 mixed cw01	Signal + SSL	0.4291	0.2277	0.2062
p80 d48 l2 mixed cw01	Signal + SSL + phenotype	0.4482	0.2267	0.2062
p120 d64 l2 block cw02	Signal + SSL	0.2925	0.2495	0.2062
p120 d64 l2 block cw02	Signal + SSL + phenotype	0.3	0.2467	0.2062

analysis type: development selected test evaluation

Table S51: Supplementary Table 9, development selected test evaluation, column block 1 of 3

config	model	selection split	AUROC	AUPRC
p80 d48 l2 mixed cw01	Signal + SSL + phenotype	selected by original-training internal validation; original held-out test evaluated once	0.6348	0.3872

Table S52: Supplementary Table 9, development selected test evaluation, column block 2 of 3

config	model	Brier score	event prevalence	Train records
p80 d48 l2 mixed cw01	Signal + SSL + phenotype	0.2542	0.2048	386

Table S53: Supplementary Table 9, development selected test evaluation, column block 3 of 3

config	model	Held-out test records	Train events	Held-out test events
p80 d48 l2 mixed cw01	Signal + SSL + phenotype	166	79	34

analysis type: dimensionality regularization

Table S54: Supplementary Table 9, dimensionality regularization, column block 1 of 4

model	AUROC	AUPRC	Brier score	event prevalence
signal + ssl64 + signal phenotype l2	0.6168	0.4016	0.2652	0.2048
Signal + SSL pca5 + signal phenotype l2	0.6286	0.3863	0.2547	0.2048
Signal + SSL pca5 + signal phenotype elasticnet	0.6279	0.3801	0.2548	0.2048
Signal + SSL pca10 + signal phenotype l2	0.6212	0.4241	0.2588	0.2048
Signal + SSL pca10 + signal phenotype elasticnet	0.6194	0.4026	0.2584	0.2048
Signal + SSL pca20 + signal phenotype l2	0.6535	0.4256	0.2566	0.2048
Signal + SSL pca20 + signal phenotype elasticnet	0.6549	0.4172	0.2557	0.2048

Table S55: Supplementary Table 9, dimensionality regularization, column block 2 of 4

model	AUROC	Train records	Held-out test records	Train events
signal + ssl64 + signal phenotype l2	0.6168	386	166	79
Signal + SSL pca5 + signal phenotype l2	0.6286	386	166	79
Signal + SSL pca5 + signal phenotype elasticnet	0.6279	386	166	79
Signal + SSL pca10 + signal phenotype l2	0.6212	386	166	79
Signal + SSL pca10 + signal phenotype elasticnet	0.6194	386	166	79
Signal + SSL pca20 + signal phenotype l2	0.6535	386	166	79
Signal + SSL pca20 + signal phenotype elasticnet	0.6549	386	166	79

Table S56: Supplementary Table 9, dimensionality regularization, column block 3 of 4

model	AUROC	Held-out test events	SSL dimensions	penalty
signal + ssl64 + signal phenotype l2	0.6168	34	64	l2
Signal + SSL pca5 + signal phenotype l2	0.6286	34	5	l2
Signal + SSL pca5 + signal phenotype elasticnet	0.6279	34	5	elasticnet
Signal + SSL pca10 + signal phenotype l2	0.6212	34	10	l2
Signal + SSL pca10 + signal phenotype elasticnet	0.6194	34	10	elasticnet
Signal + SSL pca20 + signal phenotype l2	0.6535	34	20	l2
Signal + SSL pca20 + signal phenotype elasticnet	0.6549	34	20	elasticnet

Table S57: Supplementary Table 9, dimensionality regularization, column block 4 of 4

model	AUROC	n predictors before one hot	events per predictor before one hot	PCA explained variance ratio
signal + ssl64 + signal phenotype l2	0.6168	77	1.026	
Signal + SSL pca5 + signal phenotype l2	0.6286	18	4.389	0.84
Signal + SSL pca5 + signal phenotype elasticnet	0.6279	18	4.389	0.84
Signal + SSL pca10 + signal phenotype l2	0.6212	23	3.435	0.9567
Signal + SSL pca10 + signal phenotype elasticnet	0.6194	23	3.435	0.9567
Signal + SSL pca20 + signal phenotype l2	0.6535	33	2.394	0.9953
Signal + SSL pca20 + signal phenotype elasticnet	0.6549	33	2.394	0.9953

analysis type: outcome sensitivity

Table S58: Supplementary Table 9, outcome sensitivity, column block 1 of 4

model	AUROC	AUPRC	Brier score	event prevalence
Classical signal features	0.6303	0.3366	0.2525	0.2048
Signal + SSL + phenotype	0.6168	0.4016	0.2652	0.2048
Classical signal features	0.6398	0.3882	0.2525	0.1988
Signal + SSL + phenotype	0.6744	0.4432	0.2489	0.1988
Classical signal features	0.7809	0.4773	0.2183	0.1145
Signal + SSL + phenotype	0.8006	0.487	0.1941	0.1145
Classical signal features	0.8181	0.5409	0.207	0.0904
Signal + SSL + phenotype	0.8631	0.526	0.1726	0.0904

Table S59: Supplementary Table 9, outcome sensitivity, column block 2 of 4

model	AUROC	Train records	Held-out test records	Train events
Classical signal features	0.6303	386	166	79
Signal + SSL + phenotype	0.6168	386	166	79
Classical signal features	0.6398	386	166	72
Signal + SSL + phenotype	0.6744	386	166	72
Classical signal features	0.7809	386	166	37
Signal + SSL + phenotype	0.8006	386	166	37
Classical signal features	0.8181	386	166	25
Signal + SSL + phenotype	0.8631	386	166	25

Table S60: Supplementary Table 9, outcome sensitivity, column block 3 of 4

model	AUROC	Held-out test events	outcome	outcome definition
Classical signal features	0.6303	34	neonatal risk	primary composite: cord pH <7.15 or 5-minute Apgar <7
Signal + SSL + phenotype	0.6168	34	neonatal risk	primary composite: cord pH <7.15 or 5-minute Apgar <7
Classical signal features	0.6398	33	cord ph lt 7 15	cord pH <7.15 only
Signal + SSL + phenotype	0.6744	33	cord ph lt 7 15	cord pH <7.15 only
Classical signal features	0.7809	19	cord ph lt 7 10	cord pH <7.10 only
Signal + SSL + phenotype	0.8006	19	cord ph lt 7 10	cord pH <7.10 only
Classical signal features	0.8181	15	cord ph lt 7 05	cord pH <7.05 only
Signal + SSL + phenotype	0.8631	15	cord ph lt 7 05	cord pH <7.05 only

Table S61: Supplementary Table 9, outcome sensitivity, column block 4 of 4

model	AUROC	AUPRC / test prevalence
Classical signal features	0.6303	1.643
Signal + SSL + phenotype	0.6168	1.961
Classical signal features	0.6398	1.953
Signal + SSL + phenotype	0.6744	2.229
Classical signal features	0.7809	4.17
Signal + SSL + phenotype	0.8006	4.255
Classical signal features	0.8181	5.985
Signal + SSL + phenotype	0.8631	5.821

Supplementary Table 10. TRIPOD+AI reporting checklist mapping for the manuscript

Complete source table: Supplementary_Table_10_TRIPOD_AI_checklist.csv. Rows: 52; columns: 5. The display below is the complete table.

Table S62: Supplementary Table 10

item	topic	status	reported location	comment
1	Title	Reported	Title	Identifies self-supervised CTG trajectory representation and calibrated neonatal risk enrichment in the target intrapartum CTG population.
2	Abstract	Reported	Structured abstract	Structured Background Objectives Methods Results Conclusions abstract aligned with Frontiers in Signal Processing positioning.
3a	Healthcare context and rationale	Reported	Introduction	Explains intrapartum CTG decision-support context and limitations of static or subjective interpretation.
3b	Target population and intended purpose	Reported	Introduction and Discussion	Target population is intrapartum CTG records; intended use is risk enrichment and phenotype interpretation rather than autonomous diagnosis.
3c	Known health inequalities	Partially reported	Discussion limitations	No protected sociodemographic fairness analysis was possible beyond available CTU-UHB metadata; this is treated as a limitation.

Continued from Supplementary Table 10.

item	topic	status	reported location	comment
4	Objectives	Reported	Introduction	Objective is evaluation of an open-data self-supervised CTG trajectory-representation workflow for calibrated neonatal risk enrichment.
5a	Data sources	Reported	Methods and Supplementary Table 1	CTU-UHB/CTU-CHB is primary raw-signal cohort; CTGDL-FHRMA is external representation/ domain-shift resource; UCI CTG is feature-level reference.
5b	Data dates	Partially reported	Supplementary Table 1	Public datasets are cited with access information; detailed original accrual dates depend on source metadata.
6a	Study setting	Reported	Methods and Supplementary Table 1	Intrapartum cardiotocography public data from hospital-based CTG resources.
6b	Eligibility criteria	Reported	Methods and Supplementary Table 1	Records included when raw FHR/UC and required metadata were available for the analysis pipeline.
6c	Treatments received	Not applicable	Methods	No treatment-effect modelling was performed; delivery process variables were not used as intervention exposures.

Continued from Supplementary Table 10.

item	topic	status	reported location	comment
7	Data preparation and quality checking	Reported	Methods	Signal windowing, missingness, feature extraction, train-only scaling and source-data generation are described.
8a	Outcome definition and time horizon	Reported	Methods outcome definition	Primary endpoint is cord pH <7.15 or 5-minute Apgar <7 at record level.
8b	Subjective outcome assessment	Not applicable	Methods outcome definition	Outcome uses recorded pH and Apgar metadata from the public dataset rather than new subjective adjudication.
8c	Outcome blinding	Not applicable	Methods	Retrospective public-data analysis; outcome labels were withheld from SSL pretraining.
9a	Initial predictor choice	Reported	Methods outcome and prediction models	Classical signal features and SSL embeddings were prespecified from CTG physiology and self-supervised representation design.
9b	Predictor definitions	Reported	Methods and supplementary tables	Signal features, SSL embedding dimensions and phenotype variables are described; source-data tables are included.
9c	Subjective predictor assessment	Not applicable	Methods	Predictors were algorithmically extracted from public CTG signals and metadata.

Continued from Supplementary Table 10.

item	topic	status	reported location	comment
10	Sample size	Reported	Methods and Results	Record counts, event counts and window counts are reported; limitations discuss small event count and high-dimensional model risk.
11	Missing data	Reported	Methods and Results	Median imputation, missingness features and no-missingness sensitivity are reported.
12a	Data partitioning and analysis use	Reported	Methods record split	Record-level training/test split, train-only SSL pretraining and held-out validation are stated.
12b	Predictor handling	Reported	Methods	Numeric predictors were median-imputed and standardized; categorical phenotypes were one-hot encoded.
12c	Model type and tuning	Reported	Methods and Supplementary Table 9	Transformer SSL, logistic pipelines, exploratory compact sweep and development-only selection sensitivity are described.
12d	Heterogeneity across clusters	Partially reported	Results and Discussion	Domain-shift analysis compares CTU-UHB with CTGDL-FHRMA; formal multi-centre heterogeneity was not possible.

Continued from Supplementary Table 10.

item	topic	status	reported location	comment
12e	Performance measures	Reported	Methods and Results	AUROC, AUPRC, Brier score, calibration, decision curve and bootstrap CIs are reported with rationale.
12f	Model updating or recalibration	Reported	Methods and Results	Cross-fitted Platt and isotonic recalibration are described; Platt scaling is emphasized.
12g	Prediction calculation	Reported	Methods and Code availability	Pipeline scripts are named and the versioned repository archive is available at https://doi.org/10.5281/zenodo.20485068.
13	Class imbalance	Reported	Methods and Results	Class-balanced logistic regression and AUPRC emphasis are justified under event imbalance.
14	Fairness	Partially reported	Discussion limitations	Available public metadata did not support a full fairness assessment; age and clinical stratification were limited.
15	Model output	Reported	Methods and Discussion	Outputs are risk-enrichment scores and calibrated probabilities; not intended as definitive diagnosis.
16	Training versus evaluation differences	Reported	Methods and Results	CTU train/test split and CTGDL representation-domain differences are explicitly described.

Continued from Supplementary Table 10.

item	topic	status	reported location	comment
17	Ethical approval	Reported	Declarations	Public de-identified datasets; ethics approval not applicable.
18a	Funding	Reported	Declarations	No specific funding.
18b	Conflicts of interest	Reported	Declarations	No competing interests declared.
18c	Protocol	Reported	Code availability and limitations	No formal protocol was registered; analysis scripts and source data are included for reproducibility.
18d	Registration	Reported	Code availability and limitations	Study was not registered.
18e	Data sharing	Reported	Availability of data and materials	Public data sources and derived source-data tables are listed.
18f	Code sharing	Reported	Code availability	Local scripts are identified; the GitHub repository and versioned Zenodo archive are reported in Code availability.
19	Patient and public involvement	Not applicable	Methods/Declarations	No patient or public involvement in this retrospective public-data analysis.
20a	Participant flow	Reported	Results and Figure 1	Cohort scale and train/ test records are shown.

Continued from Supplementary Table 10.

item	topic	status	reported location	comment
20b	Participant characteristics	Reported	Supplementary Table 1	Dataset scale and available record-level characteristics are summarized in supplementary files.
20c	Development versus evaluation comparison	Reported	Figure 1 and Supplementary Table 1	Train/test event balance and external domain-shift analysis are reported.
21	Number of participants and events per analysis	Reported	Results and Supplementary Table 9	Events are reported for primary and sensitivity analyses.
22	Model specification	Reported	Methods and Code availability	Architecture, feature dimensions and logistic models are described; implementation scripts are included in the GitHub repository and versioned Zenodo archive.
23a	Model performance with uncertainty	Reported	Results and Supplementary Tables 2-3	Performance estimates include bootstrap confidence intervals and model differences.
23b	Heterogeneity in performance	Partially reported	Results and Discussion	Domain-shift and missingness sensitivity are reported; subgroup fairness analysis remains limited.
24	Model updating results	Reported	Results and Supplementary Table 7	Platt and isotonic recalibration results are reported.

Continued from Supplementary Table 10.

item	topic	status	reported location	comment
25	Interpretation	Reported	Discussion	Interprets risk enrichment, phenotype physiology, calibration and external-domain boundaries.
26	Limitations	Reported	Discussion	Discusses retrospective design, model-selection sensitivity, event count, dimensionality, missingness and lack of external outcome validation.
27a	Handling poor quality or unavailable inputs	Partially reported	Methods and Discussion	Missingness sensitivity and signal-quality features are reported; deployment handling is future work.
27b	User interaction and expertise	Reported	Discussion	Model is positioned for clinician-facing decision support rather than autonomous use.
27c	Future research	Reported	Discussion	Calls for larger locked train/development/test designs, external outcome validation and prospective evaluation.

Supplementary Table 11. Transformer input, tokenization, masking, objective, optimizer and training hyperparameters

Complete source table: Supplementary_Table_11_transformer_hyperparameters.csv. Rows: 28; columns: 5. The display below is the complete table.

Table S63: Supplementary Table 11

section	parameter	selected value	sweep or fixed values	notes
Input	signal channels	FHR and UC	FHR and UC	Two-channel CTG windows were used for SSL pretraining.
Input	window length	10 minutes	Fixed	Each window contained 2400 samples per channel at 4 Hz.
Input	sampling rate	4 Hz	Fixed	Signals were resampled or represented at the project target sampling rate.
Tokenization	patch length	80 samples	80, 120	Non-overlapping patches were used in the selected model.
Tokenization	patch stride	80 samples	Equal to patch length	The selected configuration produced 30 temporal patches per channel.
Tokenization	token count	60 channel-time tokens	Derived	Thirty temporal patches by two channels.
Encoder	model dimension	64	48, 64	Latent token dimension before transformer encoding.
Encoder	transformer layers	3	2, 3	Compact encoder depth for the selected manuscript-candidate.
Encoder	attention heads	4	Fixed	Multi-head self-attention heads.
Encoder	feedforward width	4 x model dimension	Fixed	Transformer feed-forward layer width.
Encoder	activation	GELU	Fixed	Activation function in transformer feed-forward layers.

Continued from Supplementary Table 11.

section	parameter	selected value	sweep or fixed values	notes
Encoder	dropout	0.1	Fixed	Applied within the compact transformer encoder.
Masking	masking strategy	Channel masking	Mixed, block, channel	Channel masking was used by the held-out manuscript-candidate; mixed and block masking were sensitivity configurations.
Masking	mask fraction	0.25	Fixed	Fraction of patches or channel-time tokens masked in each view.
Augmentation	jitter standard deviation	0.02	Fixed	Gaussian jitter applied to paired training views.
Objective	reconstruction loss	Masked-patch mean squared error	Fixed	Computed only over masked patches.
Objective	contrastive loss	NT-Xent	Fixed	Applied to projected pooled embeddings from paired views.
Objective	contrastive temperature	0.2	Fixed	Temperature parameter for NT-Xent loss.
Objective	contrastive weight	0.2	0.0, 0.2	Weight applied to the contrastive objective in the total loss.
Projection	projection dimension	64	Fixed	Projection head output dimension for contrastive learning.
Training	optimizer	AdamW	Fixed	Optimizer used for compact candidate training.
Training	learning rate	0.001	Fixed	Learning rate for AdamW.
Training	weight decay	0.0001	Fixed	Weight decay for AdamW.

Continued from Supplementary Table 11.

section	parameter	selected value	sweep or fixed values	notes
Training	batch size	32	Fixed	Mini-batch size for SSL training.
Training	gradient norm clipping	1	Fixed	Maximum gradient norm.
Training	candidate epochs	4	Fixed	Candidate configurations in the compact sweep were trained for four epochs.
Training	selected architecture trace	12 epochs	Inspection only	The 12-epoch trace was used for training-dynamics inspection and did not determine model selection.
Downstream	record embedding	Mean pooled window embeddings	Fixed	Clean unmasked windows were encoded and averaged to record-level SSL representations.